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NBIO 101

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Misalignments of Incentive Salience as Afflictions of the Soul

The story of neurological dysfunction embeds itself into high-profile lives like that of Ernest Hemingway, whose family history included multiple suicides, including his own in 1961¹, into the deeply personal battles like those of my own family struggling with addiction and depression, and into entire communities where life expectancy has declined in recent decades². The neurobiology of dopaminergic dysfunction in addiction and depression reveals a crisis of incentive salience,³ a fundamental inability to want what sustains us, that manifests as spiritual illness in communities trapped by poverty, abuse, and trauma; typical responses through faith and community represent something beyond superstition: they are essential interventions where the brain, soul, and social structures collide. To see the connections between incentive salience and symptoms of spiritual sickness, we will begin by defining incentive salience more rigorously and exploring what happens with varying misalignments of incentive salience. Next, we will look at how chronic stress affects the human brain in ways that cause it to completely recalibrate its desires and needs in ways that are self-reinforcing. We will also look at how socioeconomic factors affect children's development by literally reshaping children's brains. Finally, we will examine how faith, communities, and practice of hope address this crisis at both neurological and

¹ Jamison, K. R. (1993). *Touched with Fire: Manic-Depressive Illness and the Artistic Temperament*. Free Press.

² Case, A., & Deaton, A. (2015). Rising morbidity and mortality in midlife among white non-Hispanic Americans in the 21st century. *Proceedings of the National Academy of Sciences*, 112(49), 15078–15083

³ Berridge, Robinson, & Aldridge (2009), Dissecting components of reward: 'liking', 'wanting', and learning, *Current Opinion in Pharmacology*, 9(1), 65–73.

spiritual levels offering pathways towards healing that consider the full scope of human spiritual needs and experiences.

When the brain misaligns its motivational systems with actual sources of wellness, attributing wanting to harmful substances rather than nourishing relationships or meaningful work, we get mental issues ranging from addiction to depression. Berridge and Robinson define incentive salience as a specific psychological process: the attribution of wanting to reward-based stimuli, which makes them both attractive and wanted. Notably, they also find that “incentive salience is a motivational 'wanting' attribute... that can be separated from hedonic 'liking' and from cognitive expectations of future pleasure.”⁴ This dissociation explains how addiction can cause the compulsive pursuit of harmful substances like drugs and alcohol even if they no longer bring us pleasure. In this same paper, Berridge posits that drug addiction results from permanent sensitization of mesocorticolimbic systems, the brain circuits that mediate dopamine-driven motivation.⁵ Repeated drug use can hyperactivate these circuits causing them to be sensitized to drug related cues even after the pleasure of actually using the drugs fades. In other words, drug addiction occurs when the wanting exceeds the liking of even pleasant drugs.

If we consider what depression is: the inability to want or pursue even basic pleasures, then we can attribute both addiction and depression to misalignments of incentive salience. Berridge notes that in addiction, incentive salience can become “detached from the object of desire and be attributed too widely among stimuli, spewing indiscriminant ‘wanting’ in directions that are inappropriate.”⁶ This means that drug related cues become associated with

⁴ Berridge, K. C., & Robinson, T. E. (2016). Liking, wanting, and the incentive-sensitization theory of addiction. *American Psychologist*, 71(8), 670–679

⁵ Berridge & Robinson, 2016, p. 675

⁶ Berridge, K. C. (2009). Wanting and liking: Observations from the neuroscience and psychology laboratory. *Inquiry*, 52(4), 378–398.

intense motivational draw, compelling drug-seeking behavior even when an addict wishes to quit. In class, Dr. Newsome highlighted that “what makes us human is ‘desire’, not DNA”. In other words, it is the capacity to want, value, and pursue things that sets us apart from biological machines. Incentive salience is the neurobiological mechanism underlying these capacities, dictating what it is these desires grasp onto. This means that when incentive salience is dysregulated, we lose our capacity to be human: we are unable to desire the things that sustain us and make us whole, and instead compulsively seek things that are destroying us. This illness goes beyond a simple chemical imbalance: it is an affliction of the soul.

When safety and stability are chronically compromised, the brain’s threat-detection systems become hyperactive.⁷ This chronic activation of stress systems can dysregulate dopaminergic reward circuits making it more difficult to generate incentive salience. In other words, when the brain is constantly under threat, its capacity to desire healthy rewards becomes compromised. When the need for safety and stability isn’t met, the mesocorticolimbic system can recalibrate to attribute intense salience from a reward to a threat. Anxiety then becomes what I am calling a form of negative incentive salience: a state where the brain becomes hypersensitive to fear.

Chronically anticipating threats and danger forces the human brain to recalibrate its dopaminergic systems of desire; the brain predicting the world to be an unrewarding and dangerous place makes it more difficult to pursue long-term goods. Sterling and Eyer first introduce the concept of allostasis to distinguish it from homeostasis.⁸ Rather than merely reacting to disturbances to maintain a homeostatic equilibrium point, allostasis considers future

⁷ Koob, G. F. (2015). The dark side of emotion: The addiction perspective. *European Journal of Pharmacology*, 753, 73–87

⁸ Sterling, P., & Eyer, J. (1988), “Allostasis: A new paradigm to explain arousal pathology,” in Fisher & Reason, *Handbook of Life Stress, Cognition and Health* pp 629–649

needs and adjusts physiological systems in advance.⁹ Sterling explains that the fundamental principle is prediction: the brain anticipates needs and mobilizes all mechanisms in concert to meet them before they arise.¹⁰ Sterling writes: “chronically anticipating a need for higher pressure, the brain mobilizes all the low level mechanisms in concert.”¹¹ This means that if the brain predicts ongoing stress, based on past experiences of violence, abuse, or poverty, it will maintain an elevated blood pressure, elevated cortisol, and altered dopamine signalling¹² not as a malfunction, but as an adaptation to better react to the world. The problem with this is that over time, this stresses and damages the body, which Sterling refers to as “allostatic load”. McEwen and Stellar specifically show that chronic allostatic load increases the risk of cardiovascular disease, diabetes, cognitive decline, and early mortality.¹³ If we consider poverty as a high stress environment where people are constantly anticipating threat, scarcity, and abuse, we can see that poverty-induced illnesses are affected by allostatic load. Evans and Kim demonstrate that an increased allostatic load leads to higher resting blood pressure, stress hormones, and BMI.¹⁴ The stress of poverty literally recalibrates physiological systems in ways that persist into the rest of an individual’s life. This allostatic framework reveals that chronic stress doesn’t just change subjective experience; it fundamentally restructures the developing brain. This reframes the question to: how does a poverty-induced allostatic load translate into altered neurocognitive development?

⁹ Sterling, P. (2012). Allostasis: A model of predictive regulation. *Physiology & Behavior*, 106(1), pp 5–15

¹⁰ Sterling 2012 pg 5

¹¹ Sterling 2012 pg 5

¹² Sterling 2012 pp 5-15

¹³ McEwen, B. S., & Stellar, E. (1993). Stress and the individual: Mechanisms leading to disease. *Archives of Internal Medicine*, 153(18), pp 2093–2101

¹⁴ Evans, G. W., & Kim, P. (2013). Childhood poverty, chronic stress, self-regulation, and coping. *Child Development Perspectives*, 7(1), pp 43–48

Hackman, Farah, and Meaney show that SES (Socioeconomic Status) is directly associated with differences in brain structure in children.¹⁵ More specifically, they find that children raised in poverty or lower SES households show measurable differences in executive function, language systems, memory systems, and stress response systems.¹⁶ Directly building on Sterling's findings, we can posit that stressors such as lower socioeconomic status and traumatic experiences create an overwhelming allostatic load that alters developing children's brains. These physical changes in developing brains are what culminate in generational cycles of poverty and sickness. I propose that the model of transmission begins with poverty: poverty subjects people to stress and trauma that alter their brains during development, and these effects often manifest in destructive behaviors such as abuse, poor emotional regulation, and impaired executive function. Those, in turn, become the stressors in their children's lives, and the cycle repeats.

Now that we have established that poverty is a condition that affects the neurobiology of the individuals subjected to it, it is important to also approach this condition as one of spiritual ailment. Spiritual wellness within the context of this paper can be defined as a connection between an individual's incentive salience or deep longings and their sources of well-being. Disconnects between incentive salience and well-being can manifest itself as meaninglessness, isolation, and despair. In the case of meaninglessness, an individual is unable to justify their suffering within a larger narrative; hope is a crucial aspect of wellness which such an individual is cut off from. Isolation happens when an individual is cut off from their community, which ends up harming their chances of survival as well as their spiritual wellness. Despair occurs

¹⁵ Hackman, D. A., Farah, M. J., & Meaney, M. J. (2010). Socioeconomic status and the brain: Mechanistic insights from human and animal research. *Nature Reviews Neuroscience*, 11(9), pp 651–659

¹⁶ Hackman, Farah, & Meaney, 2010 pp 651-659

when a person believes that their ability to change their circumstances are out of reach. I hypothesize that this connection is so tied to our survival that the body registers its loss as physical illness. These needs have profound physical effects, and we have known about them for a long time. We have spent countless years trying to attain spiritual fulfillment because it is so essential to our physical wellbeing as well as our mental wellbeing.

When we do not have this fulfillment, we find ways to attain it, and this is where faith and community are so important in rebuilding this spiritual fulfillment. In 2015, Princeton economists Case and Deaton documented “deaths of despair” in which they observed a sharp increase in the mortality and morbidity of middle aged white non-Hispanics after the 1990s.¹⁷ Rehder et al. take this one step further in their 2021 paper where they “explore theoretical models of psychopathology... that may inform our understanding of mechanisms of risk for negative mental health outcomes.”¹⁸ It is important to consider that the communities where deaths of despair concentrated were often lower-income white communities, many of them rural, including Appalachia and parts of the south. The American rural south is often associated with a profound culture of faith and worship that is a crucial unifier.

The use of faith as a weapon against despair has been present in our society for as long as humans have roamed the planet. In any culture or society humans have formed across time, faith has been nearly universal. Its power goes beyond community and promotes healing by giving people strength. My godfather used to be heavily addicted to cocaine; his life was a complete disaster, and he went into financial, spiritual, and physical ruin due to his addiction, but when he wanted to stop, he went to the local church for help and guidance. While it would be reductive to

¹⁷ Case and Deaton 2015 pp 15078–15083

¹⁸ Rehder, K., Lusk, J., & Chen, J. I. (2021). Deaths of Despair: Conceptual and Clinical Implications. *Cognitive and Behavioral Practice*, 28(1), pp 40–52

say that the church “fixed” him, the component of faith was a key contributor to his ability to overcome his addiction and place his spiritual wellbeing into the hands of something larger than himself. Not only did he feel that God was by his side even in the most difficult moments, but the community around him also supported him through the difficulty of overcoming addiction despite his irritability and sickness associated with withdrawal.

Sterling validates this idea in his later work with Platt, which asks why deaths of despair have risen so sharply in the United States but not in other developed nations. Their answer is that countries offering communal care at every stage of life from birth to old age, protect people from the isolation that drives these deaths, while Americans are left to confront the life cycle on their own.¹⁹ In other words, community is not a luxury, rather, it is a buffer against allostatic load, and this is exactly what my godfather received when he was absorbed into the church structure that helped him bear his problems and lowered the chronic stress he carried alone for years.

Hope and eschatology offer the possibility for change and play a crucial role in our spiritual wellness. If incentive salience is fundamentally predictive, attaching desire to what the brain expects to be rewarding, then hope is predicting a future worth wanting. The stories we tell about endings, and what lies beyond them, give suffering a place within a larger framework; they are reassurances that our pain is a chapter and not the whole story. This is why despair is so dangerous at an individual level: a person who expects nothing from the future has no reward to predict and thus nothing to pull them forward. Faith restores that pull.

We need an approach that combines the spiritual aspect of mental illness, the afflictions of the soul, with neurological solutions. Treating the brain alone ignores the meaninglessness, isolation, and despair that keep pulling a person back towards sickness. Treating only the soul

¹⁹ Sterling, P., & Platt, M. L. (2022). Why Deaths of Despair Are Increasing in the US and Not Other Industrial Nations. *JAMA Psychiatry*, 79(4), pp 368–374

ignores the allostatic load and dopaminergic dysregulation that make healing so physically difficult. Faith is easy to dismiss as superstition, but in our class, we discussed the power of placebo: neuroscience cannot always tell us why belief heals, only that it does, and that does not make the healing any less real. I want to be transparent with the limits of this argument. My model of intergenerational poverty is too simple: the lived reality is far more complex than a single cycle of stress and altered development can properly capture. Many of the solutions I described, such as my godfather's turn to the church, are also individual rather than structural. They do not undo the poverty and abuse that created allostatic loads to begin with. A complete answer would need to properly address these structures. Even then, my simple model is useful if it can demonstrate how the brain, the soul, and our communities are intimately bound together, and that healing any one of them means tending to all three.

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